

Question Paper

Exam Date & Time: 07-May-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

SECOND SEMESTER B.TECH. DEGREE EXAMINATIONS - APRIL / MAY 2025

SUBJECT: PHY 1071/PHY_1071 - ENGINEERING PHYSICS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Explain attenuation in optical fiber. Discuss the three main sources for attenuation in optical fibre. (4)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1B) The ratio of Einstein's coefficient (A_{21}/B_{12}) is 6.2×10^{-15} for a lasing system at 300 K. Calculate the wavelength of the radiation emitted by a laser. Calculate the rate of stimulated to spontaneous emission. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) The statement 'temperature of the atom can be reduced to nanokelvin using laser'. State whether the statement is true or false. Justify your answer. (2)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 2A) Derive the expression for Compton shift using laws of conservation of energy and momentum. Show at which scattering angle Compton shift is minimum. (4)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) X-rays of wavelength $\lambda_0 = 0.20$ nm are scattered from a block of material. The scattered X-rays are observed at an angle of 45° to the incident beam. Calculate the scattered X-ray wavelength and Compton shift. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2C) In the Davisson-Germer experiment, electrons are accelerated at 52 V potential difference. Calculate the de-Broglie wavelength of the fast-moving electrons which are ejected from Davisson-Germer experimental set-up. (2)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3A) The wave-function for a particle confined to moving in a one-dimensional box is $\psi(x) = A \sin\left(\frac{n\pi x}{L}\right)$. Find the normalised wave function and obtain energy values of the particle. (4)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3B) The frequency of photon that causes $v = 0$ to $v = 1$ transition in the CO molecule is 6.42×10^{13} Hz. Ignore any changes in the rotational energy. (4)

A) Calculate the force constant k for this molecule.

B) What is the maximum classical amplitude of vibration for this molecule in the $v = 0$ vibrational state? Use reduced mass of CO molecules as 1.14×10^{-26} Kg.

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3C) When an electron and a proton of the same kinetic energy encounter a potential barrier of the same height and width, which one of them will tunnel through the barrier more easily? Why? (2)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4A) Describe dielectric constant, susceptibility, dielectric loss, and strength. (4)

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 4B) An HCl molecule is excited to its second rotational energy level. If the distance between its nuclei is $r_0 = 0.1275$ nm, what is the angular speed(ω) of the molecule about its center of mass? (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 4C) A dielectric is placed in an electric field E_0 produces a depolarizing field E' . Is the electric field E experienced by a dipole inside the dielectric $E = E_0 + E'$? Explain. (2)

Course Outcome: "CO4" / Cognitive Level: "Application"

- 5A) With a neat diagram, explain the preparation of nanomaterials by top down and bottom-up approach give an example for each. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- 5B) Distinguish between conductors, insulators and semiconductors based on band theory. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 5C) The job of an exhaust nozzle in an aircraft is to produce thrust that helps the aircraft to maintain speed. Among metal glass, shape memory alloy, and biomaterials which material will be useful in manufacturing of exhaust nozzle and why? (2)

Course Outcome: "CO5" / Cognitive Level: "Application"

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Event Name			Subject
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025			PHY 1071-PHY:ENGINEERING PHYSICS
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	18
4	4	CO 4	6
5	5	CO 5	6
Total			50

* NA - CO Not Selected

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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	20(40%)
3	Application	10	30(60%)
4	Analysis	0	0(0%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)